

Model-Based Reflex (Semi active)

Agentic AI

Date: _____

- Agent →
 - Perceives its environment using sensors
 - Acts on that environment using effectors.

- Agentic AI →
 - Can make decisions on their own
 - acts independently

- Passive Agents → works only on what they see now (current input)
 - No memory + No learning + No goals + No planning
 - Simple reflex Agent

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Robot Vacuum

- Active Agents →
 - keep internal memory
 - Can learn from experience
 - Can plan & work towards goal.
 - Goal based + utility based agent
- Proactive Agents →
 - start actions on their own
 - based on internal goals
 - Do not wait for something outside to tell them what to do.
 - can act even before situation needs it.

quick response

- Reactive Agents → responds to what happened outside

Imp in Agentic AI

- Proactive Agents → initiate action themselves
- Deliberative Agents → plan carefully how to act

- Software Agents →
 - Purely digital
 - Virtual Sensors/ → APIs, memory.
 - effectors
 - e.g., chatbots, AutoGPT

- Embodied Agents →
 - Physical form
 - Interact with real world

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Drones, self-driving cars,
Robot arms

Sensors/ → Cameras, LIDAR
effectors

- AI is considered Agentic →
 - Autonomy
 - Goal-directed behaviour
 - Statefulness
 - Decision Authority
- A very intelligent system cannot be agentic → single GPT prompt.

Date:

Human Control in Agentic AI

Humans-in-the-loop	Humans-on-the-loop	Full autonomous Agent
- Human approval required - Safer & auditable - Healthcare + Finance.	- AI acts independently - Human monitors outcome,	- Minimal human oversight - High responsibility & risk.

Agentic AI Reasoning Frameworks

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tells agent how to think, act & improve over time.

* a) ReAct (Reason + Act) → smart loop of thinking + acting + learning

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Think (Reason) + Do (Act) + See (Observe)

(b) Reflexion (Act + Self feedback)

- keeps memory of what went right or wrong
- learn from mistakes autonomously.

* c) Plan - and - Execute (PnE)

- Plan firstly & then executes.
- Suitable for long, structured tasks.
- not good if environment changes.

* d) Tree of Thought (ToT) → outputs schema-constrained (JSON) to reduce drift.

- Explores multiple reasoning paths
- Picks strongest one before acting
- Considers alternatives, better reliability

* e) Chain of thought (CoT)

- Solves problems step by step in a straight line. (like a chain)

f) Multi-Agent systems

- Multi Agents with different roles
- Agents can collaborate, debate or specialize.
- More Robust & Creative.

g) Langchain Agents (LLM + Tool)

- Calls external tools like search engines, APIs, calculator.

Frameworks

Strengths

Best Use Case

1) ReAct	Flexible, adaptive	Dynamic Problem solving
2) Reflexion	Self-correction	Iterative tasks
3) PnE	Structured	long-term projects
4) ToT	Explores options	Strategy, Planning
5) CoT	Clear reasoning	Math, logic Problems
1) Multi-Agent	Collaboration	Complex goals
2) LLM + Tools	Resourceful	Real-world workflows

• Reasoning Frameworks → Thinking style of an agent

• Development Frameworks → Software environment that equips agents with tools, memory & actions.